Project Documentation

Full Stack Development – Project Documentation Format

Team Enrollment Confirmation – Import Data using Transform Maps (Spreadsheet)

Team Leader	SRIKAVI T
Team Member	ISWARYA S
Team Member	YOGASRI S
Team Member	KEERTHIKA L
Team Size	4

1. Introduction

- **Project Title:** Team Enrollment Confirmation – Import Data using Transform Maps (Spreadsheet)
- **Team Size:** 4

Team Members and Roles

Name	Role	Responsibility
SRIKAVI T	Team Leader	Project coordination, transform map design, final review
ISWARYA S	Team Member	Spreadsheet preparation, data source and import set configuration
YOGASRI S	Team Member	Table/form design, business rules, notifications and access control
KEERTHIKA L	Team Member	Testing, UAT, reports and documentation

2. Project Overview

- **Purpose:** To automate team enrollment confirmation by importing enrollment records from a spreadsheet (Excel/CSV) into the platform using Import Sets and Transform Maps, instead of entering every record by hand. Each imported record is validated, mapped to the target Team Enrollment table and marked as confirmed.

Key Features

- Spreadsheet upload through a Data Source (Excel / CSV) and staging in an Import Set table.
- Transform Map with field mappings from spreadsheet columns to the Team Enrollment table.
- Coalesce on Team ID + Member Email to prevent duplicate enrollments and update existing ones.
- Field mapping scripts that clean data (trim, upper/lower case, date formatting).
- Automatic enrollment status and confirmation date set during transformation.
- Enrollment confirmation notification to the team member / team leader.
- Role-based access control and reports on enrollment status.

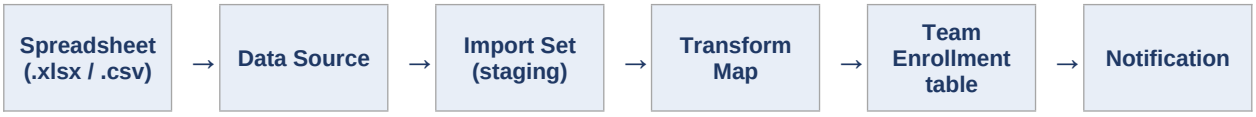
3. Architecture

Frontend

The user interface is provided by the platform's standard UI (Next Experience / Service Portal): list view and form view for the Team Enrollment table, the Load Data page for uploading spreadsheets, and a reports dashboard. Form layout, UI policies and client scripts control which fields are mandatory or read-only.

Backend

Server-side logic runs on the platform's JavaScript engine. It consists of the Data Source, the Import Set table, the Transform Map (with field maps, coalesce rules and onBefore/onAfter transform scripts), business rules, and event-driven notifications. Data is also exposed through the REST Table API and Import Set API.



Database

Data is stored in platform tables. The staging table holds raw spreadsheet rows; the target table holds validated enrollments.

Table	Field	Type	Description
u_team_enrollment_import (Import Set)	u_team_id, u_team_name, u_member_name, u_member_email, u_role, u_enrollment_date	String / Date	Raw rows loaded from the spreadsheet
u_team_enrollment (Target)	Team ID	String	Unique team identifier (coalesce field)
	Team Name	String	Name of the team
	Member Name	String	Full name of the enrolled member
	Member Email	Email	Member email (coalesce field)
	Role	Choice	Team Leader / Team Member
	Enrollment Date	Date	Date of enrollment
	Status	Choice	Pending / Confirmed / Rejected
	Confirmation Date	Date/Time	Set automatically on confirmation

Table and field names above are the names used in this project design; adjust them if your instance uses different names.

4. Setup Instructions

Prerequisites

- A ServiceNow Personal Developer Instance (PDI) or equivalent instance with admin access.
- Enrollment spreadsheet in .xlsx or .csv format with a header row.
- A modern web browser (Chrome / Edge).

- Git (optional) for storing the exported update set / application.

Installation

1. Log in to the instance as an administrator.
2. Create the Team Enrollment table (System Definition > Tables) with the fields listed in Section 3.
3. Prepare the spreadsheet with columns: Team ID, Team Name, Member Name, Member Email, Role, Enrollment Date.
4. Create the Data Source (System Import Sets > Administration > Data Sources) and attach the spreadsheet; select the sheet and header row.
5. Use Load Data to create the Import Set staging table.
6. Create the Transform Map (System Import Sets > Administration > Transform Maps) from the staging table to the Team Enrollment table.
7. Add field maps, set coalesce on Team ID and Member Email, and add the onBefore script to set Status and Confirmation Date.
8. Create the notification, ACLs and reports; export as an update set or application for backup.

5. Folder Structure

The project is a platform application, so its “folders” are application file types rather than source directories.

Client (user interface)

Component	Description
Forms / Form Layout	Team Enrollment form and list layout
UI Policies / Client Scripts	Mandatory and read-only rules, field validation
Reports / Dashboards	Enrollment status reports
Portal Page (optional)	Enrollment upload and confirmation page

Server

Component	Description
Tables	u_team_enrollment and staging import set table
Data Source	Excel / CSV data source definition
Transform Map	Field maps, coalesce rules, onBefore/onAfter scripts
Business Rules / Flow	Enrollment confirmation logic
Notifications / Events	Confirmation email
ACLs / Roles	Access control for enrollment data

6. Running the Application

9. Open System Import Sets > Load Data and choose the Data Source / upload the spreadsheet.
10. Click Submit to load rows into the Import Set staging table.
11. Click Loading Data > Transform (or open the Transform Map and choose Transform) to run the map.
12. Open the Team Enrollment list to view the imported and confirmed records.
13. Check the Import Set Rows / Transform History for inserted, updated and ignored counts.

For automated runs, a Scheduled Import can be created on the Data Source to execute the load and transform at a set frequency.

7. API Documentation

The platform exposes the enrollment data through the REST Table API and the Import Set API. All calls require authentication (Section 8).

Method	Endpoint	Description / Parameters
POST	/api/now/import/ u_team_enrollment_import	Insert a row into the Import Set; body: team, member and role fields. Transform Map runs automatically.
GET	/api/now/table/u_team_enrollment	List enrollments. Params: sysparm_query, sysparm_limit, sysparm_fields
GET	/api/now/table/u_team_enrollment/ {sys_id}	Get one enrollment record
PATCH	/api/now/table/u_team_enrollment/ {sys_id}	Update fields (for example status)
DELETE	/api/now/table/u_team_enrollment/ {sys_id}	Delete an enrollment record

Example request – POST /api/now/import/u_team_enrollment_import

```
{
  "u_team_id": "TEAM001",
  "u_team_name": "Team Alpha",
  "u_member_name": "ISWARYA S",
  "u_member_email": "member@example.com",
  "u_role": "Team Member"
}
```

Example response

```
{
  "import_set": "ISET0010001",
  "staging_table": "u_team_enrollment_import",
  "result": [{
    "transform_map": "Team Enrollment Map",
    "table": "u_team_enrollment",
    "display_value": "TEAM001",
```

```
    "status": "inserted"
  }]
}
```

8. Authentication

- Users sign in with instance credentials (local or SSO); browser access uses a session cookie.
- REST clients authenticate with HTTP Basic Auth over HTTPS or OAuth 2.0 bearer tokens.
- Authorization uses roles and ACLs: only users with the enrollment admin role can load data and run transform maps; team members have read access to their own records.
- Import and transform actions are logged in the Import Set and Transform History tables for auditing.

9. User Interface

Screenshots of the main screens are listed below. Insert the captured images in the placeholders.

[Insert screenshot: Load Data page with spreadsheet upload]

[Insert screenshot: Transform Map field mapping]

[Insert screenshot: Team Enrollment list view]

10. Testing

Testing follows a functional and user acceptance approach performed on the instance:

- Unit testing of each field map and transform script with a small sample spreadsheet.
- Data validation testing: blank fields, invalid emails, wrong date formats and duplicate rows.
- Coalesce testing: re-importing the same file must update, not duplicate, records.
- Notification and ACL testing with different user roles.
- User Acceptance Testing (see the separate UAT report).

Tools: Import Set / Transform History logs, Script Debugger, the REST API Explorer and browser developer tools.

11. Screenshots or Demo

[Insert screenshot: Import set rows and transform results]

[Insert screenshot: Confirmed enrollments report]

Demo link: [Add project demo video / GitHub link here]

12. Known Issues

- Spreadsheet column headers must match the expected names, otherwise field mapping fails.
- Very large files should be split into batches to avoid transform timeouts.
- Dates must use a consistent format (for example YYYY-MM-DD) or a conversion script is needed.
- Error rows are only visible in the Import Set Row Errors list; there is no automatic error email yet.

13. Future Enhancements

- Automatic scheduled imports from a shared location or attachment on a record.
- Error summary email to the team leader after every import.
- Self-service portal page for members to confirm or decline enrollment.
- Dashboards for enrollment trends and team-wise participation.
- Integration with an approval workflow and external HR/college systems.